

VALENCIA AP, Spain

WMO: 082840

Lat: 39.485N

Lon: 0.475W

Elev: 62

StdP: 100.58

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.8	2.1	-9.0	1.8	7.9	-6.2	2.2	9.0	14.1	15.6	12.4	14.6	2.4	280	0.413

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.4	33.7	20.7	32.0	21.3	31.0	21.6	24.8	29.1	24.1	28.5	23.5	28.1	5.0	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.3	18.2	27.4	22.8	17.7	27.2	22.0	16.8	26.7	75.5	29.1	72.8	28.6	70.4	28.0	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.8	9.3	7.9	DB	-2.0	38.5	1.7	2.4	-3.2	40.3	-4.2	41.7	-5.2	43.0	-6.4	44.7	(4)
			WB	-3.5	25.5	1.8	1.4	-4.8	26.5	-5.9	27.3	-6.9	28.1	-8.3	29.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	17.8	10.8	11.3	13.5	15.7	19.1	23.0	25.8	26.1	23.0	19.3	14.2	11.3	(6)
		DBStd	6.08	3.03	3.02	2.95	2.49	2.59	2.32	1.64	1.73	2.38	2.66	3.11	2.95	(7)
		HDD10.0	72	25	18	4	1	0	0	0	0	0	4	21		(8)
		HDD18.3	1058	235	197	153	85	21	1	0	1	19	127	219		(9)
		CDD10.0	2919	49	55	113	171	281	392	489	500	389	290	131	59	(10)
		CDD18.3	864	0	1	3	6	44	142	231	242	141	51	4	1	(11)
		CDH23.3	7643	1	4	35	60	313	1192	2292	2428	1013	283	21	1	(12)
		CDH26.7	2532	0	0	4	7	70	361	841	922	272	53	3	0	(13)
(14)	Wind	WSAvg	3.2	3.6	3.6	3.6	3.5	3.1	3.0	3.1	3.0	2.9	2.7	3.2	3.2	(14)
(15)	Precipitation	PrecAvg	449	34	30	35	30	34	22	6	15	69	79	50	44	(15)
		PrecMax	834	153	106	118	90	143	132	34	50	260	366	202	168	(16)
		PrecMin	221	0	0	0	1	0	0	0	0	0	0	0	0	(17)
		PrecStd	164	36	27	33	24	37	27	8	13	69	98	55	44	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	22.3	23.8	26.9	27.3	31.3	34.9	35.4	37.0	33.9	30.6	25.9	21.8		(19)
		0.4% MCWB	13.1	13.4	15.0	15.8	17.5	20.3	19.7	20.4	19.8	18.5	16.0	14.0		(20)
		2% DB	20.0	21.0	23.8	25.1	28.4	31.9	33.0	33.7	31.0	28.0	23.1	19.9		(21)
		2% MCWB	12.2	12.4	13.8	14.7	17.4	19.6	21.3	21.3	20.4	18.1	14.9	13.1		(22)
		5% DB	18.1	19.0	21.7	23.1	26.3	30.1	31.7	32.0	29.4	26.2	21.2	18.2		(23)
		5% MCWB	11.4	11.5	12.7	13.9	16.6	19.9	21.7	22.3	20.7	18.2	14.4	12.1		(24)
		10% DB	16.6	17.2	19.8	21.6	24.8	28.7	30.6	30.9	28.1	24.8	19.7	16.9		(25)
		10% MCWB	10.7	10.7	12.2	13.8	16.5	19.6	21.9	22.5	20.6	18.0	13.7	11.5		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	14.4	14.6	16.3	17.4	20.5	24.0	25.2	25.7	24.5	21.7	18.0	15.3		(27)
		0.4% MCDB	20.0	21.3	23.9	23.4	26.3	28.5	29.7	30.1	28.3	25.0	22.3	19.8		(28)
		2% WB	13.2	13.4	14.7	16.2	19.2	22.6	24.5	25.1	23.4	20.6	16.7	14.1		(29)
		2% MCDB	18.0	18.9	21.1	21.7	24.7	27.6	29.0	29.5	27.4	24.5	20.7	18.1		(30)
		5% WB	12.3	12.4	13.8	15.3	18.4	21.7	23.8	24.5	22.6	19.8	15.7	13.1		(31)
		5% MCDB	16.6	17.3	19.5	20.7	23.6	27.0	28.5	28.9	26.7	23.8	19.4	17.0		(32)
		10% WB	11.4	11.6	13.1	14.6	17.7	20.9	23.2	23.9	21.9	19.0	14.8	12.2		(33)
		10% MCDB	15.5	15.9	17.9	19.8	22.8	26.2	28.0	28.3	26.2	23.1	18.5	15.9		(34)
(35)	Mean Daily Temperature Range	MDBR	10.7	10.8	11.1	10.6	10.4	10.2	9.5	9.4	9.6	9.9	10.1	10.4		(35)
		5% DB MCDBR	12.7	13.6	14.4	13.6	12.9	12.5	11.6	11.6	11.5	12.3	11.9	11.8		(36)
		5% DB MCWBR	7.0	7.2	6.9	6.2	5.4	5.0	5.1	5.3	5.2	5.7	6.3	6.7		(37)
		5% WB MCDBR	10.5	11.8	12.1	11.6	10.5	10.0	9.5	9.2	9.4	10.0	9.9	10.2		(38)
(40)	Clear-Sky Solar Irradiance	taub	0.306	0.328	0.373	0.391	0.417	0.437	0.435	0.441	0.419	0.379	0.334	0.311		(40)
		taud	2.521	2.479	2.370	2.336	2.294	2.277	2.299	2.286	2.339	2.442	2.514	2.526		(41)
		Ebn at Noon	870	894	881	885	865	844	842	827	823	823	827	835		(42)
		Edh at Noon	75	90	111	123	131	133	129	128	114	93	76	70		(43)
(44)	All-Sky Solar Radiation	RadAvg	2.36	3.18	4.40	5.50	6.50	7.44	7.39	6.31	4.86	3.55	2.49	2.04		(44)
		RadStd	0.27	0.36	0.43	0.41	0.47	0.29	0.24	0.28	0.30	0.31	0.28	0.17		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	-2.30	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	-1.35	+0.41	N/A	N/A	N/A	N/A	N/A	+49

Nomenclature: See separate page